

Assignment M5

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Abstract—Over the course of the M assignments, I will be investigating the task of searching for local events, with a focus on the Meetup.com website. Meetup.com’s search page currently has two main page views and several filters, but there are some use cases that are difficult to accomplish (for example, searching for certain types of events).

1 QUALITATIVE EVALUATION

1.1 Evaluation Overview

For this evaluation, I gave a survey using the PeerSurvey tool on a paper prototype. *Appendix 4.1: Paper Prototype for Qualitative Evaluation (from M3)* shows a picture of the prototype. Twenty five participants took part, and those participants were mostly (23 of the 25) recruited from classmates in this course, while the other two were family members. The survey went generally well. I did implement some feedback from early survey results. One respondent recommended I make one of the questions not required, so part way through I removed that restriction, so it was not required for the rest of the participants. The only organizational change I would make next time would be to give fewer questions. I noticed that other surveys I completed from peers were not as long, and I think that I may have overwhelmed participants with the number of questions.

1.2 Raw Results

The full survey results are in *Appendix 4.2: Qualitative Evaluation Survey Results*.

Participants were mostly in the 18-29 (36%) or 30-39 (52%) age groups. They mostly lived in urban (44%) or suburban (52%) areas. When they thought back to a recent year that was unaffected by COVID-19 restrictions, they searched for meetups approximately 0 times (24%), 1-3 times (44%), or 4-6 times (24%) during that year. They actually attended a meetup approximately 0 times (28%) or 1-3 times (60%) in that year.

When asked how often they would use different filters, the Genre filter was popular (72% would use frequently or very frequently), as was the Location filter

(92% would use frequently or very frequently), the Date(s) filter (88% would use frequently or very frequently), and the Time(s) filter (76% would use frequently or very frequently). However, the Number of Attendees filter was less popular, with only 40% reporting they would use it frequently or very frequently.

Participants were also asked about the look and feel of the app. 92% reported that they agreed or strongly agreed that the interface looked easy to use, and 84% agreed or strongly agreed that it looked enjoyable to use. 96% agreed or strongly agreed that the results were displayed in a way that was easy to understand, and 96% agreed or strongly agreed that they were confident they could find a meetup relevant to their interests using this interface.

Participants were asked some free form questions as well. When asked what their typical goal for searching for a meetup, common answers included to find people with similar interests, to learn a new skill, or to participate in a particular activity. When asked what filters they recommended adding, common answers included cost (free or paid) and age group or median age. Participants liked that the interface was simple and easy to use. Common dislikes include the zip code filter (they suggested other location filters), clutter, and lack of more granular filtering or sorting (e.g. a start and end time). The final suggestions and comments ranged considerably, but many reflected earlier topics, like needing ways other than zip code to determine location.

1.3 Analysis of Feedback

The relative popularity of each of the filters was what I expected. I thought that Number of Attendees would be less popular than the others but the others would be perceived as useful, and that was correct. There were some filters that were recommended that I never thought of, for example the cost (free versus paid), or age groups. These are both great ideas and I am surprised they never occurred to me. Another suggestion was to filter based on if an event was a one-off or part of a series, which is also a great idea.

A very common piece of feedback that was delivered in several different ways was to change the location filter. I appreciate how many different ideas people had; to use a map, to have a dropdown of different neighborhoods instead of zip codes, to show distance from current location, or be able to search by address. This is something that I would have to think about and iterate on in future design cycles to find a solution that is more helpful for users but does not add

clutter to the design.

Some feedback I received was contradictory, which was interesting. For example, many participants said they liked how clean and minimalist the design is, while others said it was too cluttered. This is not actionable, but I thought it was interesting how differently the same design can be perceived by different people.

Overall, participants seemed to enjoy the interface but also gave some really good ideas for how it could be improved, which I appreciated.

1.4 Future Iterations

One change I would love to make based off of this feedback is to add in the suggested filters, such as age group, cost, and whether an event is recurring or not. I would also add in sorting and keyword searching, both of which make sense and can be helpful for narrowing down the list. Another participant recommended a "calendar" view instead of a list view; I think it would be nice to have two tabs, one for the list and one for a calendar. These are all changes that could easily be added (do not require a full redesign) but would bring a lot more functionality to the design. The general design of the filters and list seem to be easy to use, so I would keep those for future iterations.

2 PREDICTIVE EVALUATION

2.1 Evaluation Overview

This is a cognitive walkthrough for a textual prototype, which is shown in full in *Appendix 4.3: Textual Prototype for Predictive Evaluation (from M3)*. As explained in M4, I am walking through the four scenarios that I did in the needfinding stages so I can be consistent in how the interface is evaluated.

2.2 Execution

Before even executing one of the searching or filtering tasks, users need to find the right location. The main app view is a map that is currently centered around a default location (e.g. San Francisco) or around the user's current location if they have shared it, with a blue circle indicating where they currently are. Using the principle of consistency, the user would be able to understand how to change the location in the map to their current location based on how it is done in similar applications: either by allowing the app to use their location or by

searching for their city or zip code in the search bar at the top. They then would use direct manipulation of the map to either move the map around or zoom in and out of the map.

Clicking on an icon to learn more about it is consistent with other applications as well. Also due to the tolerance principle, users will feel safe to click around different options without worrying about making any major mistakes. After the icon is clicked, a modal appears with a summary of more information. There are two signifiers on this modal: an "X" to close the modal, and a "More" button that will take them to a new page with all of the information about the event. On this new page is another signifier, a "back" button, that will take them back to the original page. These buttons are also consistent with other applications so users will feel comfortable knowing which buttons to press.

2.2.1 Task 1: Finding a particular genre of event (for example, a running meetup)

The icons on a map use the concept of consistency and mapping to help the user understand what the icons mean. For example, an icon of a running person is both consistent with other applications (like emojis in text messaging applications) and maps to the real-world concept of people running. This helps the user understand that an icon of a running person might signify a running event.

However, other events may not have icons that are as consistent or as clearly mapped. There also may be too many different types of categories. It is not clear in the textual prototype how many types of categories there are, so for example there may just be one icon for all exercise meetups or there may be a different icon for each type of exercise meetup. If there is one for every type that may be confusing, and you would need very granular icons. However, if there is only one icon for each general category, then it may not be clear to click on a running person to find a soccer meetup, for example.

2.2.2 Task 2: Finding events over a range of dates (for example, this weekend)

There is a date picker at the top of the screen. This is consistent with other applications and how they do filtering, so a user will be able to apply that principle to know to click on that filter to sort by date. Also, because of the principle of discoverability, they can see all of the filters at once to try them out. The date filter is by default set to two weeks, so users know that they can always change the filter back if they make a mistake, which is an example of the

tolerance principle.

2.2.3 Task 3: Finding the next time a particular group is meeting up (for example, a specific book club)

This particular task is actually difficult using this interface, and the user might not know what to do. It may require that the user has some outside knowledge about the event. For example, if the user already knows when and where the book club meets, they can move the map to that location (or search for that location in the search bar), then they can use the date picker to select the appropriate dates. If they only know one of those pieces of information, they can try filtering by that first and then clicking around to find the right meetup. If they do not know when or where, but they know it is a book club, they might start clicking on book icons due to the principle of mapping (it would make sense to the user that the book icon maps to a book club). After clicking on each book club, they can read the summary view and evaluate if that is the club they are looking for or not.

2.2.4 Task 4: Combining multiple filters (for example, finding a running meetup next weekend)

This particular situation is a combination of task 1 and task 2, so the same observations apply for this task. The order of operations also does not matter, so due to the tolerance principle users can feel free to try different things out to find their result. For example, they can change the dates first and then look for the correct icon for a running meetup, or they can click on different running meetup icons and then check the date.

3 EVALUATION SUMMARY

3.1 Needfinding About the User

One question that arose from the qualitative evaluation was what is the main obstacle for users in attending the meetups that they search for? It is clear from the survey that people searched for more meetups than they actually attended, but it is not clear the reason why. It could be due to a simple reason like time constraint, or it could be because they were not finding good results, or it could be because of a bad experience at a meetup that held them back from attending future events. This could be an interesting thing to ask more about, because the

current design may not be giving users the information that they need to fulfill their goal of attending a meetup.

I would also like to learn more about users' previous experience with attending meetups and what made a meetup a positive or negative experience. Those factors might be good things to highlight on the meetup website, or perhaps could be ways to filter events. Some users mentioned in the survey that cost or age group has been a negative aspect of meetups for them in the past, and I wonder if there are other factors that they would want to know about before attending an event. If users have more positive experiences at the meetups they attend, it is reasonable to expect that they would search for more meetups to attend in the future.

3.2 Design Alternatives

One good design alternative would be to combine the two designs that were evaluated, the paper and the textual prototype. The textual prototype includes a map, and according to the survey, users suggested including a map. Users also suggested being able to search for addresses and not just zip codes, which the textual prototype also includes. Therefore, a good design alternative would be to include the map view as a tab separate from the list view that is in the paper prototype. This would allow users to use the view that they find the easiest to browse through.

Another design alternative would be to ask users some basic questions in their profile about the target demographic of their ideal meetup. That would help users automatically filter out groups not in their target age range, for example.

3.3 Revisions to the Prototype

For the next iteration, I would make the changes that I have listed earlier in this document for the paper prototype: adding the age range filter, adding the cost filter, adding the recurring/single event filter, and changing the way locations are searched for. For the location filter change, I would change the zip code filter to a general location filter so users can input an address and a mileage range. These are small changes that do not fundamentally alter the design. The overall design was overwhelmingly reviewed as positive by the participants of the survey, so I want to keep the main aspects of the design while still incorporating the advice of the participants.

For the next level of fidelity, I would create a wireframe of the prototype. The main design is the same, so it doesn't make sense to evaluate it again at this level of fidelity with only a few small filters different. The wireframe will make it easier for users to read and understand the different sections. It also might make it more clear and less "clunky" to those who felt that the design wasn't simple enough. It is too early to incorporate fonts and colors, but I do think that some specificity in the wireframe (such as relative difference in font size) will make it easier for users to evaluate the look and feel and decide if it is clean and minimalist enough and easy to use.

3.4 Evaluation Plan

After this stage, I still do not think that my designs would be ready for a true empirical evaluation. Searching for meetups is a lot more nuanced than I thought when I originally started this assignment series, and users are thinking about a lot of different aspects of events when they are looking for meetups. Therefore, I would want to do another qualitative evaluation at the end of this design round to understand if users feel that their needs are being met not only for searching for events, but this time also for finding events that they will attend. Previously, I was asking more about browsing behavior, like how are users browsing for events. I was not focusing on what makes events stand out to users and encourages them to attend in real life. Therefore, the next round needs to do a qualitative evaluation to understand if users can find events they actually will go to. If the qualitative evaluation is successful, then the round after this one can be used for an empirical evaluation.

4 APPENDIX

4.1 Paper Prototype for Qualitative Evaluation (from M3)

This is shown in Figure 1, below.

4.2 Qualitative Evaluation Survey Results

For all results below, the question is in the table caption, the possible answers are the columns, and the percentages are the percentage of respondents who chose that answer (out of 25 respondents). Free response questions have each response as its own row.

Meetup.com

FILTER BY Show/Hide

Genre

☒ Running
☒ Book Club
☒ Soccer

Location

☒ 78704
☒ 78705

From 7/4/21 To 7/6/21

Date(s)

From 5:00 PM To 8:00 PM

Time(s)

At Least 5

Number of Attendees

FILTER

RESULTS

ATX Women Run 7/5/21 at 6 PM in 78704	Running	11 Attending
Historical Fiction Book Club 7/6/21 at 7 PM in 78705	Book Club	6 Attending
Science Fiction Book Club 7/5/21 at 8 PM in 78704	Book club	22 Attending
Co-Ed Soccer League 7/4/21 at 5 PM in 78704	Soccer	74 Attending
Amstn Soccer Club 7/4/21 at 7 PM in 78705	Soccer	32 Attending

Figure 1—A paper prototype of the filters and results page

Table 1—Select your age

Under 18	18-29	30-39	40-49	50-64	65+
4%	36%	52%	8%	0%	0%

Table 2—Which of the following best describes the area you live in?

Urban	Suburban	Rural
44%	52%	4%

Table 3—Please think back to a recent year that was unaffected by COVID-19 (e.g. 2019). Approximately how many times in that year did you search for a local meetup?

0 times per year	1-3 times per year	4-6 times per year	7-10 times per year	11+ times per year
24%	44%	24%	8%	0%

Table 4—Please think back to a recent year that was unaffected by COVID-19 (e.g. 2019). Approximately how many times in that year did you attend a local meetup?

0 times per year	1-3 times per year	4-6 times per year	7-10 times per year	11+ times per year
28%	60%	12%	0%	0%

Table 5—When searching for a meetup, what is your typical goal?

Free-Form Response
See what was available in my area.
Look for a time and place to meetup
i dont use meetup
find people to run with
to find a specific activity to participate in
specific meetup to learn things
Finding people to play music with, in my area.
I had just moved to a new city. I wanted something free or cheap to do regularly to make some friends.
hiking
Meeting people.
N/A. I don't attend meetups.
I haven't search for local meetup, but if I were to, my goals would be the time, location, and interest.
learning / self education
Meet new people and/or find a group with which I can do an activity that I enjoy
find like minded individuals who enjoy the same activity
to network with ppl with similar interests as mine
find groups with similar interest
To find one that peaks my interest at the time
meet new friends
finding something im interested in
I aim at meeting people with same interests
find interesting meetup
Finding people to network with, I found Facebook groups to be better for more social events
Friendship
Church group

Table 6—How often do you think you would use the Genre filter when searching for a meetup?

Very Frequently	Frequently	Occasionally	Rarely	Never
24%	48%	20%	4%	4%

Table 7—How often do you think you would use the Location filter when searching for a meetup?

Very Frequently	Frequently	Occasionally	Rarely	Never
68%	24%	4%	4%	0%

Table 8—How often do you think you would use the Date(s) filter when searching for a meetup?

Very Frequently	Frequently	Occasionally	Rarely	Never
44%	44%	4%	4%	4%

4.3 Textual Prototype for Predictive Evaluation (from M3)

For this idea, the main app view (both for mobile and web - smartphones, laptops, etc.) would be a map. If the user allows sharing their location with the app, then the map would be centered around their current location with a blue circle indicating where they currently are. If the user does not allow sharing their location, the map would be defaulted to a particular area (for example, many apps default to San Francisco) and then users can search in a search bar at the top of the screen for a particular city or zip code.

Once users have navigated to the correct location in the map, then they can start using their fingers to drag and zoom in/out of the map. There will be an icon at each location on the map that has an event coming up in the next two weeks by default. There is a date picker at the top of the screen so that users can adjust this date range to be whatever they want (to show only events happening today, for example). The icon will correspond to the general genre of the event (a book icon for book club, for example).

If the user clicks on icon, a summary view pops up showing the event's name, location, date/time, and a brief description. If they click the X on the summary view popup, it closes the view. If they click the "More" button on the summary view popup, it navigates to a new page with more information (this new page

Table 9—How often do you think you would use the Time(s) filter when searching for a meetup?

Very Frequently	Frequently	Occasionally	Rarely	Never
32%	44%	16%	4%	4%

Table 10—How often do you think you would use the Number of Attendees filter when searching for a meetup?

Very Frequently	Frequently	Occasionally	Rarely	Never
16%	24%	32%	20%	8%

will have a back button so users can return to the map easily). If the same location has more than one event in the next two weeks, then the summary view would only show event titles and date/times and users would have to click the "More" button to see the descriptions.

Users can favorite certain venues or certain locations (like their home or work) so that those locations are highlighted with a star on the map for future reference. This view does not currently allow any filters other than location and date so that the map is the main focus, though that may change in the future.

Table 11—Are there any filters you would recommend adding to this interface? (Not required, so not all participants responded)

Free-Form Response
Perhaps if you could follow users on the app and see if any of the users you follow are attending (ex: filter to at least 1 person I follow is going to this meet up). Also day-of-week.
Free versus paid meetup
no
make this an optional question
cost (free vs paid)
no
Experience level at the activity
Filter by price. Also filter by median age and gender of RSVP'ed participants. If I had a dollar for every event I thought would be interesting but then I show up and everyone is 3x my age or 1/3 my age...
In the location filter, I would like to filter it based on the distance of my location e.g. within 5 miles.
Participant ages (or age categories) if possible/available
No
good as is
Events that are a part of a series or regular schedule. For example weekly meetups versus one off events
Activity type
Age group

Table 12—This interface looks easy to use.

Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
48%	44%	8%	0%	0%

Table 13—This interface looks enjoyable to use.

Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
40%	44%	12%	4%	0%

Table 14—Assuming that a meetup relevant to my interests exists, I am confident that I can find it by using this interface.

Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
32%	64%	4%	0%	0%

Table 15—The way the results are displayed in the interface is easy to understand.

Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
44%	52%	0%	4%	0%

Table 16—What do you like about this interface?

Free-Form Response
I like the filters. Its simple and seems easy to use.
It helps me find meetups based on my interest
useful and easy to use
contains necessary filters to be useful
minimalistic, well organized, clear search filters
clean and simple
Easy to understand how to use it
Shows all results in one list
easy to use with all relevant filters
The groupby feature.
Simple and easy to use
The interface is very simple to use.
Lots of customization features
It's simplicity and usability
It's simplicity to achieve the goal of finding a meetup
It seems very simple and intuitive to use.
Simplicity and discoverability
Everything
results listed below
simple, straightforward
A really simple and intuitive interface.
easy to use and remember
Filtering feels clear and useful. The ability to add multiple zip codes is also nice
Variety
Clear filters

Table 17—What do you not like about this interface?

Free-Form Response
Filtering by zip is a bit awkward. Perhaps it should be an x mile radius from my current location.
I'd like to see what other people are also searching for
nothing
a bit clunky
maybe could use some multimedia in the final iteration, allow the events to attract your attention
n/a
For results, zip codes can be misleading, for instance something might have a similar zip code to mine and be on the other side of my city. I'd rather have a visual like a map to compare how far away meetups are.
Too boxy, too much whitespace. Also I have never once filtered by date. I prefer to look at everything from now and scroll down a bit through the next week. I rarely plan more than 1-2 weeks ahead.
NIL
The interface is quite busy.
It's difficult for me to associate postal code to the actual location (I only remember the postal code where I live).
All the interest are displayed on the screen. If the admin happen to add few hundreds interests, the screen would be clutter. My think to use searchable dropdown.
Doesn't seem to be a way to sort results by name, date, or time
I think it's good.
There are more modern date and time entries than relying on separate fields.
None
Filter button should not be in dead center
Nothing
keywords searching
filters take up a lot of space
The interface is good and simplistic but an additional field of address for the meetup would greatly help with the usability of the system.
good as is
I want the time range to allow for just a start or end time. For example if I need to be home by 11:30 but don't care when the event starts.
Nothing comes to mind
Nothing

Table 18—Do you have any suggestions or other comments about this interface?

Free-Form Response
No
Add close results in another spot that might not match exactly what I want but maybe i would be interested in.
none
re-think the design to make it less clunky, while maintaining the usability.
n/a
rename Genre to "topics"?
Being able to save meetups from the results could be nice. Or expanding a result to see a little more detail.
Show prices. I want free activities but 95% of listings are just ads for local bars and paid classes. The real issue is Meetup's model of charging the person making the listing.
auto-fill / smart recommendation based on what I have browsed and attended before
Display results in a calendar rather than a list.
Provide an option to show meetups on a map based on where they will be held.
In terms of the timing of the day, I have different availability for different days... so would like to do the search based on that. On the interface, the From and To times user can select, they will be applied on all the dates. So in that case I need to change the dates again and again for different individual dates as I had to enter time again and again. Not sure how you would handle that in the design.
no
Add ages (or age groups) of participants. Some people may use it to meet people for dating.
no
Instead of zipcodes, use city or neighborhood names, which are more recognizable than zipcodes.
Filter button should not be in dead center. Also, I hope date fields are calendar interface.
No suggestions. I think it is great
more choices for searching
no
If you could add the location details that would be a great addition to the interface.
add some icons
I think the currently attending information is useful. I just don't really believe in it being an accurate measure anymore.
Excellent
No